

5 (a) (i) $1.5 T = 15 \text{ ms}$, hence, $T = 10 \text{ ms}$ [1]

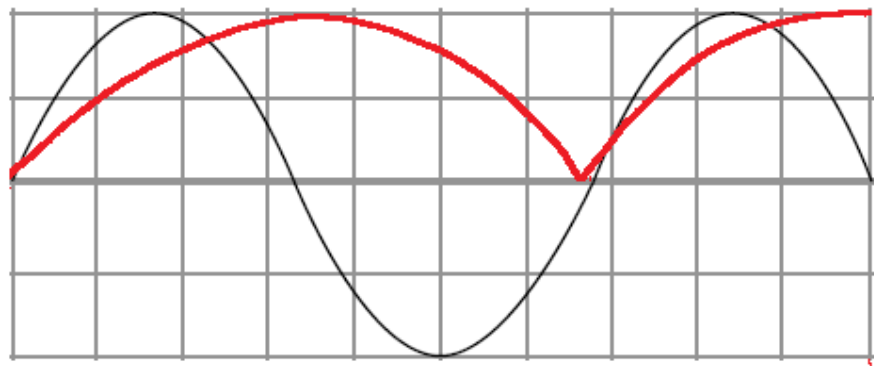
$$f = \frac{1}{T} = \frac{1}{0.010} = 100 \text{ Hz} \quad [1]$$

[Accuracy mark: Penalise 1 m if student get $f = 98.8 \text{ Hz}$]

$$V_o = 3.0 \text{ V} \quad [1]$$

$$V_{\text{rms}} = \frac{V_o}{\sqrt{2}} = 2.12 \text{ V} \quad [1]$$

(ii)



- (b) (High frequency) Alternating current that flows through the induction coil produces changing magnetic field / flux / flux density experienced by the metal cookware. [1]

According to Faraday's law, e.m.f. will be induced in the metal cookware due to the changing magnetic field. [1]

(Eddy) Currents will thus be induced in the metal cookware and heat is dissipated in it, which heats up the food within. [1]

The induction cooker works only on metal cookware because current / emf cannot be induced in non-metallic cookware such as wood or ceramic [1]